| Values of the Random Variable $X$ | 0 | 1 | 2 | Total |
|---|---|---|---|---|
| Number of elements in inverse images | 1 | 6 | 3 | 10 |

**Illustration 11.2**

A batch of 150 students is taken in 4 buses to an excursion. There are 38 students in the first bus, 36 in second bus, 32 in the third bus, and the remaining students in the fourth bus. When the buses arrive at the destination, one of the 150 students is randomly chosen.

Suppose that $X$ denotes the number of students on the bus of that randomly chosen student. Then $X$ takes on the values 32, 36, 38, and 44.

**Example 11. 4**

Two balls are chosen randomly from an urn containing 6 white and 4 black balls. Suppose that we win ₹ 30 for each black ball selected and we lose ₹ 20 for each white ball selected. If $X$ denotes the winning amount, find the values of $X$ and number of points in its inverse images.

**Solution**

The possible events of selection are (i) both balls may be black, or (ii) one white and one black or (iii) both are white. Therefore $X$ is a random variable that take the values,

$X$ (both are black balls) = ₹ 2(30) = ₹ 60

$X$ (one black and one white ball) = ₹ 30 – ₹ 20 = ₹ 10

$X$ (both are white balls) = ₹ 2(– 20) = – ₹ 40

Therefore $X$ takes on the values 60,10, and – 40.

**Note :** The inverse image of 40 is $\{b_1b_2, b_1b_3, b_1b_4, b_2b_3, b_2b_4, b_3b_4.\}$

| Values of the Random Variable $X$ | 60 | 10 | – 40 | Total |
|---|---|---|---|---|
| Number of elements in inverse images | 6 | 24 | 15 | 45 |

$\binom{6}{0}\binom{4}{2}$ $\binom{6}{1}\binom{4}{1}$ $\binom{6}{2}\binom{4}{0}$ $\binom{10}{2}$

**Illustration 11.3**

A coin is tossed until head occurs.

The sample space is $S = \{H, TH, TTH, TTTH, \ldots\}$.

Suppose $X$ denotes the number of times the coin is tossed until head occurs.

Then the random variable $X$ takes on the values $1, 2, 3, \ldots$

**Illustration 11.4**

Suppose $N$ is the number of customers in the queue that arrive at a service desk during a time period, then the sample space should be the set of non-negative integers. That is $S = \{0, 1, 2, 3, \ldots\}$ and $N$ is a random variable that takes on the values 0, 1, 2, 3, …

**Illustration 11.5**

If an experiment consists in observing the lifetime of an electrical bulb, then a sample space would be the life time of electrical bulb. Therefore the sample space is $S = [0, \infty)$. Suppose $X$ denotes the lifetime of the bulb, then $X$ is a random variable that takes on the values in $[0, \infty)$.

**Illustration 11.6**

Let $D$ be a disk of radius $r$. Suppose a point is chosen at random in $D$. Let $X$ denote the distance of the point from the centre. Then the sample space $S = D$ and $X$ is the random variable that takes on any number from 0 to $r$. That is $X(\omega) \in [0, r]$, for $\omega \in S$.



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